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TREATMENT OF THE HYPERVENTILATION SYNDROME WITH BISOPROLOL: A PLACEBO-CONTROLLED CLINICAL TRIAL

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Abstract—The hyperventilation syndrome (HVS) can be regarded as a form of panic disorder associated with a relative increase in sympathomimetic tone, the effects of which can be counterbalanced by beta-adrenoceptor blockade. The efficacy of the beta-blocker bisoprolol was investigated in a double-blind placebo-controlled randomised crossover trial involving 60 patients from 17 general practices. Following a single-blind placebo prephase, patients who met the inclusion criteria were randomised to treatment with either 5 mg bisoprolol or an identical-looking placebo tablet once daily for three weeks. They were then crossed over to the other treatment arm. At the end of each treatment phase the number of hyperventilation attacks and the severity of symptoms were assessed and side effects recorded. The number of attacks decreased from 4.04 per week at baseline to 3.52 with placebo and to 1.26 with bisoprolol. The decrease of attacks with bisoprolol was significant ($p < 0.05$) compared to baseline and placebo. The severity of the complaints improved from 29 (scale 0 to 64) at baseline not significantly to 26 with placebo and significantly ($p < 0.05$) to 15 with bisoprolol. No serious side effects were reported. Five milligrams of bisoprolol once daily is effective and safe in the maintenance of symptom reduction in patients with the hyperventilation syndrome.

INTRODUCTION

Hyperventilation is defined as an increase in the frequency and depth of respiration in excess of the requirements of the metabolic demand, leading to excessive loss of carbon dioxide. Prolonged over-breathing causes respiratory alkalosis, which in turn produces profound biochemical, neuromuscular, and neurovascular changes. The final result is a series of complaints and symptoms known as the hyperventilation syndrome (HVS) [1].

In 1871, Da Costa first described the symptoms of a syndrome characterised by repeated panic attacks. More recently the *Diagnostic and Statistical Manual of Mental Disorders (DSM-III)* has created the category of panic disorders as a diagnostic entity in its own right, distinguishing it from generalised anxiety disorder [2]. There is a considerable overlap between panic disorders and the HVS [3–5]. The reported incidence of HVS varies from 6% to 11%, depending on the patient population [6]. The diagnosis of the HVS can be made by biological criteria (Table I) and a specific questionnaire

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Table I.—Criteria for the diagnosis of hyperventilation syndrome (HVS)

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1. Ventilation rate at rest >10 L/min and abnormal (irregular with frequent gasping).
 2. End-tidal PACO_2 is low (<33.8 mmHg)
 3. Increase of PACO_2 results in a decrease and not an increase in ventilation and/or breathing frequency (in 40% of patients).
 4. After one minute of voluntary hyperventilation, there is no:
 - gradual change in the PACO_2 or the breathing rate
 - return of PACO_2 to the preprovocation level (0.75/3 min) (this is called the flywheel phenomenon).
 5. Blood gas analysis shows the base excess to be < -2: chronic respiratory alkalosis.
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For the diagnosis HVS, in either its acute or chronic form, at least three of the above criteria must be fulfilled.

[7] (Table II). The Nijmegen questionnaire is internationally validated, shows a high correlation ($r = 0.82$) with the biological criteria [8, 9], and is suitable for the evaluation of the success of therapeutic intervention.

Therapy for acute HVS consists of increasing the dead space by, *e.g.*, breathing into a bag or any procedure that increases arterial carbon dioxide levels. For maintenance therapy of chronic cases, behavioural therapy [10], sedatives, and some antidepressants have been used with success [11, 12]. Beta-blockers possess anxiolytic and ventilatory depressing qualities [8]. In the literature, the success of beta-blockade in the maintenance treatment of HVS is reported as casuistic reports [13], as controlled study in volunteers [14, 15], or in single-blind trials with the nonselective beta-blocker propranolol [16]. Bisoprolol is a beta-1-selective blocker, with a moderate lipophilic property, which enables it to pass the blood-brain barrier [17]. The elimination half-life is 12 hours, which allows for a once-a-day dosage. Bisoprolol has been shown to lack bron-

Table II.—The Nijmegen questionnaire consists of a list of complaints and a five-point scale. A total score of 22 or more gives a positive indication of hyperventilation with a specificity of 95% and a sensitivity of 91%*

Symptoms during attacks	0 Never	1 Seldom	2 Sometimes	3 Often	4 Very often
1) Chest pain					
2) Faster or deeper breathing					
3) Dyspnea					
4) Inability to breath deeply enough					
5) Tight feeling in or around the chest					
6) Tenseness					
7) Pounding heart					
8) Numbness around the mouth					
9) Stiff fingers or arms					
10) Cold hands or feet					
11) Tingling in the fingers					
12) Bloated feeling in abdomen					
13) Dizziness					
14) Haze before the eyes					
15) Confusion or sense of losing normal contact with surroundings					
16) Anxiety					

choconstrictive effects in volunteers and asthmatic patients [18, 19]. The purpose of the present study was to determine if bisoprolol is effective in the treatment of the hyperventilation syndrome.

METHODS

Subjects

Sixty patients from 17 general practices in the Netherlands were enrolled in the study. All patients were diagnosed as suffering from the HVS due to a functional mental disorder, with at least three attacks per week, defined by specific complaints and a score of more than 22 points on the Nijmegen questionnaire. Excluded were patients whose hyperventilation was caused by organic disorders such as diabetes mellitus, angina pectoris, asthmatic lung diseases, or allergic disorders. Also excluded were patients with mental disorders (other than panic attacks) who required treatment. Patients with excessive use of tobacco (>25 cigarettes per day), coffee (>10 cups per day), and alcoholic beverages (>3 units per day) were also excluded. Concomitant therapy with benzodiazepines, beta-blockers, antidepressants, or nonsteroidal anti-inflammatory drugs were not allowed.

Procedures

This study was designed as a double-blind placebo-controlled randomised crossover trial. At entry to the study, history, physical examination, heart rate, bloodpressure, ECG, and blood samples for chemical pathology were taken, the Nijmegen questionnaire was completed, and diaries were handed to the patients and explained. The patient had to enter the number of hyperventilation attacks, the severity of the attacks, and the side effects each day. Throughout the study, heart rate and blood pressure were measured by a second person (study nurse) in order to maintain the blinding of the study. At the commencement of the study, patients were given placebo (single-blind) for two weeks (Table III). After two weeks of placebo treatment, heart rate and blood pressure were recorded, the adverse events checked, and the diaries collected, controlled, and discussed with the patients. Compliance was checked by counting the returned medication, and the questionnaire score was recorded. This second score was used as baseline score. Patients in whom the score remained positive (>22) were entered into the double-blind phase of the study and were randomly assigned to treatment with either bisoprolol 5 mg or an identical-looking placebo tablet daily. After three weeks of treatment, heart rate and blood pressure were again recorded, adverse events were checked, diaries were collected and discussed, and the questionnaire was again completed. The patient was then administered the second drug for a second three-week treatment period. There was no interim placebo period because the treatment period was long enough to exclude any carry-over effect. At the end of this second active treatment period, the same procedure was followed, blood pressure and heart rate were recorded, adverse events were checked, diaries were collected and discussed with the patients, and evaluation of complaints recorded in the questionnaire was once again scored. Compliance was checked by counting the number of returned tablets.

The study was approved by the Ethical Committee of the Zuiderziekenhuis in Rotterdam, and all patients gave their written or witnessed oral informed consent to participate.

Table III. — Trial design

		Treatment periods		
		Baseline	1st period	2nd period
		placebo	Bisoprolol placebo	Bisoprolol placebo
Time in weeks	2	3	3	3
Visit number	1	2	3	4
Questionnaire	x	x	x	x
Heart rate	x	x	x	x
Blood pressure	x	x	x	x
Compliance		x	x	x
Diary		x	x	x

Table IV. — Characteristics

	Full evaluation	Dropouts	Total
Number of patients	51	9	60
Men	23	3	26
Women	28	6	34
Mean age (years)	38.1	32.7	38.0
Smokers	31	5	36
Nonsmokers	20	4	24
History of hyperventilation	18	1	19
Concomitant medication	12	1	13
Syst. Blood pressure (mmHg)	129	127	128
Diast. Blood pressure (mmHg)	80	82	81
Heart rate (beats/min)	84	78	83
Coffee consumption	49	7	56
Alcoholic beverages/day	1.9	2.2	2.0

Statistical analysis procedure

Bias due to a considerable between-subject variability of symptoms in the hyperventilation syndrome is eliminated by the use of a crossover design study. In crossover trials, each subject serves as his own control. However, a problem is the so-called treatment-by-period interaction. Carry-over effect tends to minimise differences between two consecutive treatment periods. On the other hand, the frustrating experience of an inactive agent during the first treatment period has a negative impact on the judgement of the second period. In order to exclude possible carryover effects we used the nonparametric Koch procedure [20] for a two-way crossover design with the Mann-Whitney *U* test for the evaluation of the average scores. Means and standard deviations of both treatment groups were calculated. Patients were considered to respond to the treatment if the number of attacks decreased to one or fewer per week and the score on the questionnaire decreased to below 20. The responders and nonresponders were summarised and analysed by the chi-square test. The difference was considered significant when $p < 0.05$.

RESULTS

Sixty patients were enrolled in the trial: 26 men and 34 women, mean age 38 years. Further characteristics are given in Table IV. Nine patients did not enter the double-blind phase as they no longer had a score (>22) on the questionnaire at the second visit (baseline). At baseline the patients reported an average number of 4.04 ± 1.25 attacks of hyperventilation per week. The number of attacks decreased significantly ($p < 0.05$) to 3.51 ± 2.11 with placebo and significantly ($p < 0.001$) to 1.26 ± 1.25 with bisoprolol, the difference between the two treatment regimes was highly significant ($p < 0.0001$). The severity of the complaints, measured as the results of the questionnaire score, decreased from 29 ± 3.2 at baseline to 25 ± 1.8 with placebo (not significantly) and decreased significantly ($p < 0.01$) with bisoprolol from 31 ± 4.1 to 15 ± 3.4 . With bisoprolol the improvement of the severity of the attacks scores was significantly ($p < 0.05$) greater than with placebo.

These results demonstrate that with bisoprolol, 48 (94%) patients responded to the treatment and no longer had attacks of hyperventilation meeting the criteria of hyperventilation, although in the diary of 9 patients the arousal of attacks that did not develop into full attacks was reported. With placebo treatment, 16 (32%) no longer had attacks, and three reported in their diaries arousal of attacks. The difference between the two-drug regime was significant ($p < 0.05$) (Table V). Eight patients stopped hyperventilating under placebo in the first treatment period and remained free of symptoms during the following treatment period with bisoprolol. Of the patients who ceased hyper-

Table V.—Patients with hyperventilation according to the questionnaire ($N = 51$)

	Nonresponders		Responders	
	<i>N</i>	(%)	<i>N</i>	(%)
First treatment period				
Placebo	17	(33)	8	(16)
Bisoprolol	1	(2)	25*	(49)
Second treatment period				
Placebo	18	(35)	8	(16)
Bisoprolol	2	(4)	23*	(45)
Both treatment periods combined				
Placebo	35	(68)	16	(32)
Bisoprolol	3	(6)	48*	(94)

* $p < 0.001$.

ventilating with bisoprolol in the first period, 17 out of 25 (68%) patients started hyperventilating again with placebo. All patients had normal blood pressures at the start of the trial. There was no influence of either treatment on blood pressure. Placebo did not influence heart rate, but bisoprolol reduced the heart rate by ~5 to 7 beats per minute (Table VI) which was significant ($p < 0.001$). There were no adverse events reported other than those complaints mentioned in the questionnaire.

DISCUSSION

Prior to this study, no double-blind, randomised, placebo-controlled crossover study of beta-blockers in hyperventilation syndrome has been reported. In this syndrome, the interindividual differences are so large that for a parallel group-designed trial many patients would be required. In a crossover-designed trial, the frequently self-limiting character of the syndrome might cause problems in the interpretation of the results. To avoid this problem, we included only patients who had suffered for at least six weeks with more than three attacks of hyperventilation per week. The score list of the Nijmegen questionnaire provides us with a useful diagnostic tool [9] and allows for solid statistical tests in the follow-up. In daily practice, the medical history of the patients is the most important parameter. The daily entries of incidence of attacks in the diary were the most important parameter for evaluation of the efficacy of the drug regimens. The reports of the patients in their diaries of the sensations of the arousal of hyperventilation attacks that did not develop into full-blown attacks has to be interpreted as a treatment effect, namely, the increased ability of the patients to cope with their symptoms instead of panicking.

In this study we found that for both the number of attacks and the severity of the

Table VI.—Influence of treatment on heart rate and blood pressure

	Baseline	Placebo	Bisoprolol
Mean systolic blood pressure (mmHg)	126	127	125
Mean diastolic blood pressure (mmHg)	79	80	80
Mean heart rate (beats/min)	79	77	72*

* $p < 0.05$.

complaints there was a significantly better response with bisoprolol than with placebo. While being treated with bisoprolol 94% of the patients improved. Some of them (35%) resumed hyperventilating after the end of the active treatment period with bisoprolol, indicating that maintenance treatment is required. In general practice, beta-blockers are commonly used for the treatment of hypertension, and their place in the treatment of hyperventilation is modest. One of the reasons might be the difficulty of the differential diagnosis of dyspnoeic attacks, which includes asthmatic attacks and cardiac insufficiency. In such cases nonselective beta-blockers, like propranolol, are contraindicated. bisoprolol, a selective beta-blocker, has been shown to lack bronchoconstrictive effects in volunteers and asthmatic patients [18, 19], and its efficacy and safety in congestive heart failure patients had been proved [22]. In addition to proven efficacy in the maintenance of symptom reduction in our study, this safety profile has the advantage that the physician can feel more secure in giving this specific beta-blocker to dyspnoeic patients.

The reports in the literature on successful treatment of hyperventilation always refer to beta-blockers that are lipophilic and can pass the blood-brain barrier, like propranolol and metoprolol [16, 17]. Bisoprolol is a moderate lipophilic substance that passes the blood-brain barrier and has central effects like those of propranolol [20].

Treatment with Bisoprolol did not lead to significant side effects. The treatment did not interfere with the patients' normal daily activities. One patient successfully ran the Rotterdam Marathon while taking bisoprolol.

Although the acute hyperventilation syndrome may be successfully treated by increasing dead-space breathing and behaviour therapy, drugs are frequently prescribed for maintenance therapy. Benzodiazepines and some antidepressants, especially those with serotonergic properties, have been reported as successful [11, 12], but benzodiazepines are considered less suitable for long-term treatment. Antidepressants may have several unpleasant side effects. Both benzodiazepines and antidepressants may induce dependence and interfere with concentration and normal daily activities like driving a car. Beta-blockers are well-accepted drugs, widely used for indications like hypertension, and are without a socially negative image.

In conclusion, the results of this study demonstrate that bisoprolol a "beta-1-selective beta-blocker" is effective in the maintenance of symptom reduction in patients with hyperventilation syndrome.

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